Governing the Possible: A Vector-based Analysis of the 'State of Exception' in the EU CSA Regulation through the Lens of Louise Amoore

Joel Constantin Arenz

March 1, 2026

1 Introduction & Theoretical Framework

The European Union's proposed regulation on combating Child Sexual Abuse Material (CSAM) has ignited a fierce debate over the boundaries of privacy, security, and civil liberties. This essay analyzes the discourse surrounding this regulation through the lens of Louise Amoore's "politics of possibility" (Amoore, 2013) and Giorgio Agamben's "state of exception" (Agamben, 2005). By utilizing semantic vector analysis, this study tries to surface and examine the underlying power dynamics within the "Regulation laying down rules to prevent and combat child sexual abuse" (CSA Regulation).

The initial proposal, introduced by the European Commission in May 2022, suggested mandatory "detection orders" requiring communication service providers to scan all messages for illicit material (European Commission, 2022). This move prompted immediate conflict. Only two months later, the European Data Protection Board (EDPB) and Supervisor (EDPS) raised concerns regarding the necessity and proportionality of such measures (EDPB & EDPS, 2022). Following a year of intense discussion, the Committee on Civil Liberties, Justice and Home Affairs (LIBE) released a report significantly amending the text to better protect fundamental digital rights (European Parliament, 2023).

In the context of the CSA Regulation, the protection of minors serves as a "moral shield" for a transformative form of governance — one that replaces judicial evidence with algorithmic suspicion and trades the certainty of the past for the preemption of a "dark future." By invoking the protection of the vulnerable to justify the suspension of fundamental rights, the Commission enacts what Agamben defines as the *state of exception*: a condition where the sovereign suspends the law to supposedly protect it.

Amoore identifies this power as becoming increasingly calculative, delegating the sovereign's decision-making to the "politics of possibility" and data-driven preemption. Here, the "exception" is no longer a temporary moment of crisis but is embedded into the very algorithms of detection.

To empirically map this transition from judicial certainty to preemptive possibility, this study employs a vector-based discourse analysis. If, as Amoore suggests, sovereign moves are delegated to the "calculative," then the analysis of this power must engage with the mathematical structures of legislative language. By

representing the documents in a multi-dimensional vector space using embeddings, this research tries to capture the latent semantic positions in the CSA debate.

A custom *Amoore Axis* anchored by the conceptual poles of preemptive Exception and Legal Norms is constructed to locate each institution's discursive position. In doing so, the methodology mirrors the algorithmic processes it critiques, using data science to expose the shifting boundaries of the digital state of exception.

Given the European Commission's unique "Right of Initiative," it acts as the primary agenda setter - the architect of the exception. Conversely, the LIBE Committee and the EDPB/EDPS occupy roles of legal oversight and reaction.

Out of this context of the CSA Regulation the hypothesis is defined as:

> *The European Commission's proposal will be positioned closest to the preemptive pole, while the LIBE Committee report and the EDPB/EDPS joint opinion will be positioned closer to the legal norms pole, with the EDPB/EDPS opinion being the closest to the legal norms pole.*

This paper contributes a replicable embedding-based operationalization of Amoore's framework and applies it to a high-profile case of EU legislative discourse, offering insights into the evolving dynamics of power and governance in the digital age.

2 Methodology

To map the power dynamics within the CSA Regulation's legislative process, this study analyzes three pivotal documents: the initial proposal by the European Commission (European Commission, 2022), the joint opinion of the EDPB and EDPS (EDPB & EDPS, 2022), and the LIBE Committee report (European Parliament, 2023). Each document underwent preprocessing to remove structural formatting (e.g. page numbers, headers) that lacks semantic value.

The raw text is segmented into "chunks" of approximately 500 words. To preserve semantic continuity, boundaries were defined by sentences rather than strict word counts, allowing for a 50-word overlap between adjacent chunks. This ensures that the context of a legal argument is not cut off mid-sentence.

Due to the varying lengths of the documents, with the LIBE report accounting for 56% of the total text, the dataset is downsampled to match the volume of the shortest document (the EDPB/EDPS opinion). A fixed seed is used during this random downsampling to ensure computational reproducibility. This resulted in a balanced corpus of 711 chunks per institution (2,133 total), allowing for a fair comparison of "Amoore-Scores" without the statistical bias that unequal document sizes may introduce.

To transform these text chunks into analyzable data, the Sentence-BERT (SBERT) model all-MiniLM-L6-v2 (Hugging Face, 2020) is deployed. This transformer-based model is optimized for semantic search and performs well, while not being too extensive with the computational power needed. It encodes the semantic meaning of each 500-word chunk into a dense vector of 384 dimensions.

In this high-dimensional space, the relative position of vectors represents their semantic relationship: vectors pointing in a similar direction indicate similar thematic content. By treating the legislative discourse

as a spatial distribution of these vectors, we can mathematically interpret the proximity and therefore the ideological alignment between the different institutional actors.

The initial similarity between documents is measured using Cosine Similarity. Cosine Similarity calculates the cosine of the angle between two vectors. This metric effectively answers, "How aligned are these two institutional positions?" on a scale of -1 (diametrically opposed) to +1 (identical).

To bridge this mathematical space with Amoore's theory, a custom Amoore-Axis within the existing vector space is constructed. This axis is defined by two conceptual poles:

1. The Preemption Pole: Anchored by terms emphasizing risk detection, algorithmic assessment, and proactive intervention.

2. The Legal Norms Pole: Anchored by terms emphasizing legal certainty, due process, and judicial oversight.

These poles are established by calculating the mean embedding of ten theory-driven anchor terms for each category (see Appendix 5.4). The Amoore Axis is the directional vector resulting from the subtraction of the "Legal Norms" mean vector from the "Preemption" mean vector. Each document chunk is then projected onto this axis. A positive Amoore-Score indicates alignment with preemptive exception, while a negative score suggests an orientation toward legal norms.

To ensure the selection of anchor terms is not arbitrary, a sensitivity analysis is performed. Gemini 3.0 is used to generate a broad pool of candidate terms that are then embedded using the embedding model. From these candidates the ten terms with the highest cosine similarity to the core concepts of "preemption" and "legal norms" were selected. Comparing the results of these empirically-derived anchors against the theory-driven anchors allows for a validation of the institutional positions, ensuring the findings are grounded in the semantic reality of the model rather than researcher bias.

3 Results & Analysis

The following analysis reports the semantic similarity and Amoore-Score results, and then assesses whether the observed institutional ordering supports the hypothesis. The similarity of the documents, calculated via cosine similarity, establishes a baseline for discourse proximity in the CSA Regulation Figure 4. These scores reflect the similarity in vocabulary and framing style at a document level. Higher pairwise values indicate that institutions utilize a similar linguistic style, whereas lower values signify greater distance in framing. The LIBE report demonstrates a cosine similarity of 0.853 when compared to the joint opinion of the EDPB and EDPS. This is lower than the 0.936 similarity observed between the LIBE report and the original proposal by the European Commission. Despite being released last, the LIBE report remains closer to the initial proposal, potentially adopting parts of the established framing and shifting the discourse back toward the position of the Commission. The joint opinion remains the most distinct, with a cosine similarity of 0.858 relative to the Commission, indicating that its position is less aligned with the proposal than that of the LIBE report.

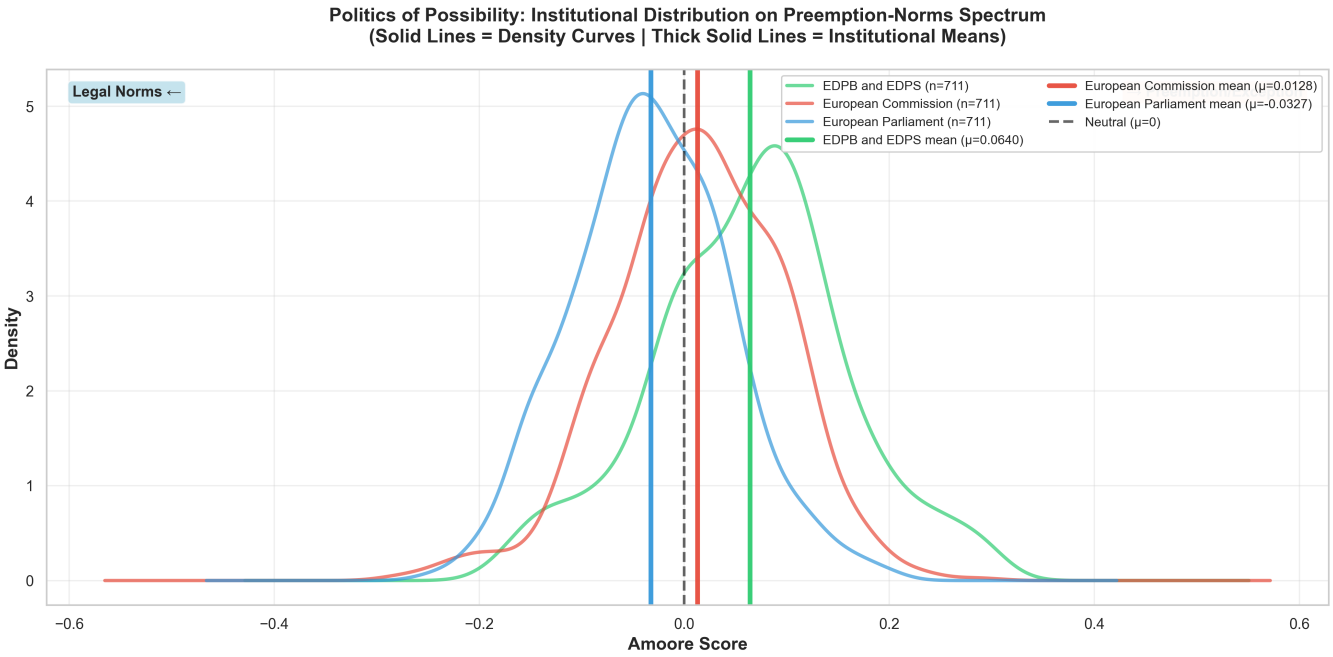

Figure 1: Institutional distribution on the Amoore-Axis using anchors selected by theory.

This analysis allows for an understanding of the discourse as a vector space, where institutional positions are represented by the distribution of Amoore-Scores across document chunks. Although the joint opinion diverges more in similarity, all three institutions remain relatively close, sharing a common discourse space. This makes the Amoore-Axis especially useful for revealing differences in normative meaning that are not immediately visible through the document similarity measure.

Figure 1 shows the Amoore density plot as the distribution of chunk-level scores on the Norms/Exception axis for the three institutions. As a robustness check, the density analysis is reproduced using empirically selected anchors (Figure 2). The institutional ordering remains identical for both iterations. The LIBE Committee (European Parliament), with a mean score of -0.0515, is positioned furthest toward the Norms pole. The European Commission's tendency toward a pole changes slightly with different anchor configurations. With theory-driven anchors, the mean is 0.0128, while empirical anchors result in a mean of -0.0086. In both cases, the confidence interval does not cover the zero effect (Figure 3). Positioned furthest toward the Exception pole in both cases are the EDPB and EDPS, with mean scores of 0.0640 and 0.0539.

The consistency across anchor selection methods reinforces the robustness of the observed institutional positioning within the semantic space. However, regarding the hypothesis, these results provide only partial support. While the LIBE report is indeed positioned toward the Norms pole, the European Commission does not occupy a significantly more preemptive position. Instead, it occupies an intermediate position. The placement of the EDPB and EDPS is the most unexpected finding. As advisory bodies that typically advocate for legal safeguards, a position closer to the Norms pole is expected, but they are positioned closest to the Preemption pole.

These positions, while only partially supporting the hypothesis, can be read through Amoore's frame-

work. The LIBE Committee's orientation toward the Norms pole fits its institutional role as a guardian of civil liberties, suggesting that the parliamentary amendment process acted as a normative corrective to the original proposal. The Commission's intermediate position may reflect a tension between two roles: on the one hand, it is the author of the preemptive framework; on the other, it must ground its proposal in existing EU law. In Amoore's terms, the exception is not introduced as a dramatic break from the legal order but is embedded within its familiar language. The positioning of the EDPB and EDPS is the most surprising finding. As bodies tasked with defending data protection rights, one would expect them closer to the Norms pole — a result that calls for closer methodological examination.

A critical result of this study is that institutional mean scores are all positioned close to the center of the Amoore axis, and the distributions show significant overlap. This suggests that the texts are not strongly polarized, and raises the question of whether the method fully captures the intended theoretical concepts. Several factors may explain this pattern. First, all institutions utilize a similar EU legal-policy vocabulary, which reduces semantic distance even when political positions differ. Second, although embedding models capture semantic meaning beyond only word frequency, the single Amoore-Axis may still merge *topic* with *normative stance*: a passage that critically evaluates algorithmic surveillance inhabits a similar region of the embedding space as one that advocates for it, because both engage with the same conceptual domain. This may account for the unexpected positioning of the EDPB and EDPS. As bodies that must engage extensively with the concepts of preemptive surveillance in order to critique them, they may be drawn toward the Exception pole not because of their normative position, but because of the subject matter they address. Additionally, the 500-word chunking strategy may average out strong local arguments, and the use of single documents per institution may further limit visible differences. Taken together, the findings may indicate a real convergence in the language used by EU bodies or specific limits in the current operationalization of the theoretical framework.

4 Conclusion

This study examined whether the legislative discourse surrounding the EU CSA Regulation reflects a shift from legal norms to a "state of exception" and preemptive governance as theorized by Louise Amoore. By embedding legislative text into a high-dimensional semantic vector space and constructing a custom Amoore-Axis, the analysis mapped the positions of the European Commission, the LIBE Committee, and the EDPB/EDPS. The results reveal that while measurable differences in institutional positioning exist, all three bodies cluster near the center of the axis with significant distributional overlap. This lack of strong polarization suggests that the discourse is not defined by a clear struggle between opposing ideologies, but rather by a shared linguistic space. This raises critical questions about whether the observed results stem from methodological limitations or represent a convergence in language, where the logic of preemption has become an integrated part of standard institutional communication.

The unexpected positioning of the EDPB and EDPS toward the preemption pole further highlights the complexity of this semantic space. Rather than indicating a support for the suspension of legal norms, this positioning likely reflects a discursive necessity within the digital state of exception. In order to critique

the regulation, advisory bodies must engage extensively with the technical and calculative vocabulary of risk, and algorithmic detection. This suggests that the "politics of possibility" may be so dominant that even the actors defending fundamental rights are forced to adopt the language of the sovereign exception.

A central direction for future research concerns the configuration of the Amoore-Axis itself. The current operationalization collapses several distinct conceptual tensions present in Amoore's framework — preemption versus legal norms, certainty versus possibility, algorithmic governance versus judicial oversight — into a single dimension. Decomposing these into separate, theoretically grounded axes could reveal whether institutions diverge on some dimensions while converging on others, offering a more differentiated mapping of the discursive space. The configuration of the poles, and therefore the axis, is a key element in translating Amoore's theory into a replicable methodological approach, and exploring alternative axis constructions would clarify how sensitively the method captures the theoretical distinctions it aims to operationalize. Equally important is validating the axis's ability to distinguish between the topic of a passage and its normative stance. A qualitative analysis of chunks with extreme scores would clarify whether the model captures evaluative positioning or only thematic proximity. Finally, expanding this methodology to a broader set of documents or different legislative cases would help determine if the observed semantic convergence is specific to the CSA Regulation or indicative of a broader trend in how the European Union governs digital risk. Ultimately, using the tools of data science to critique algorithmic governance provides a replicable framework for monitoring the evolving boundaries of the digital state of exception.

References

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5 Appendix

5.1 Anchor Terms Sensitivity Analysis

Figure 2 shows the institutional score distributions using empirically selected anchors as a sensitivity check of the Amoore-axis results.

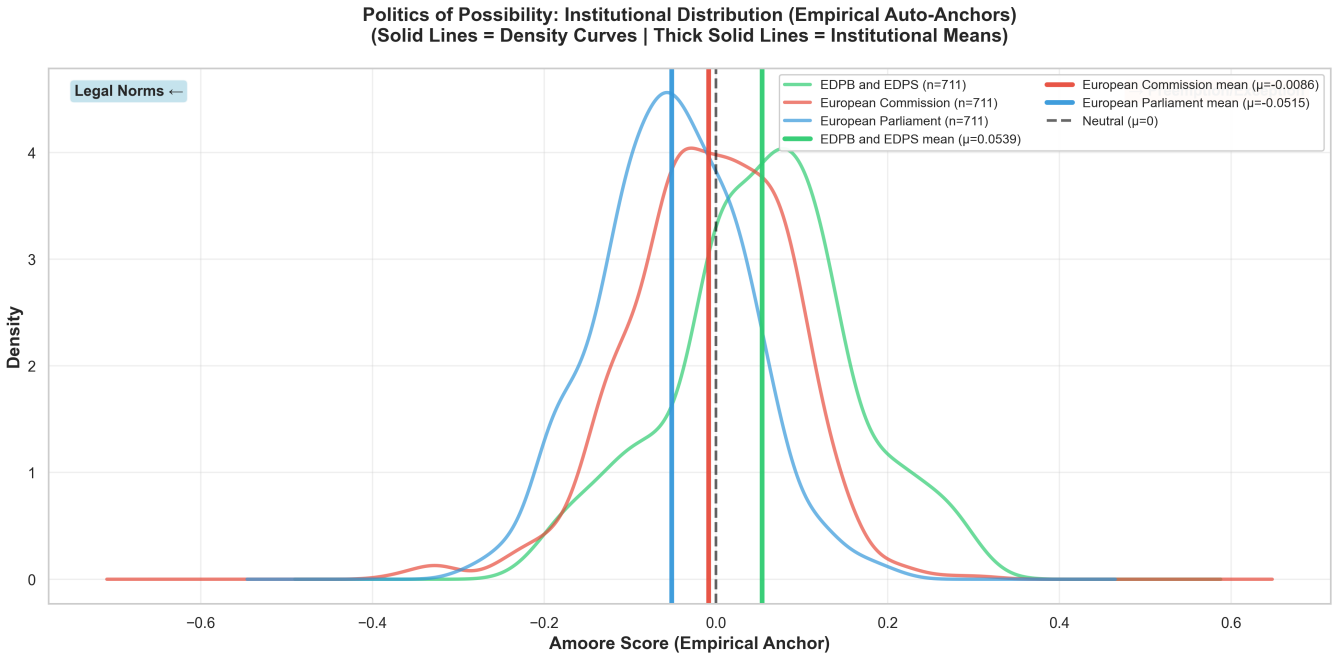

Figure 2: Institutional distribution on the Amoore-Axis using empirically selected anchors for sensitivity analysis.

5.2 Amoore Scores and Confidence Intervals

Figure 3 reports institutional mean Amoore scores with 95% confidence intervals, comparing manual and empirically selected anchors.

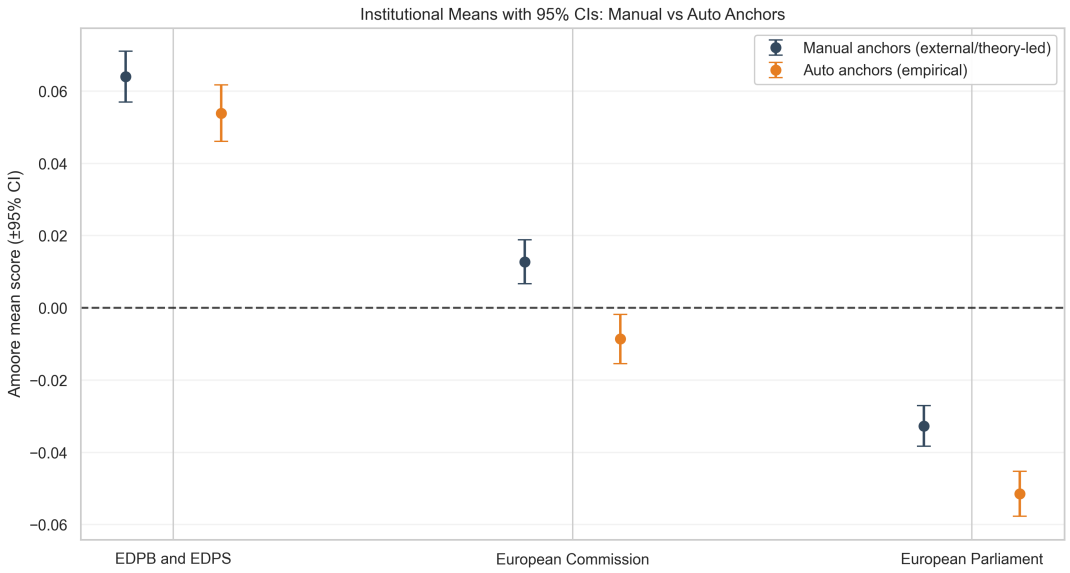

Figure 3: Institutional Means and their Confidence Interval (95%)

5.3 Document Similarity Heatmap

Figure 4 presents the institution-level cosine-similarity heatmap to summarize overall semantic proximity across documents.

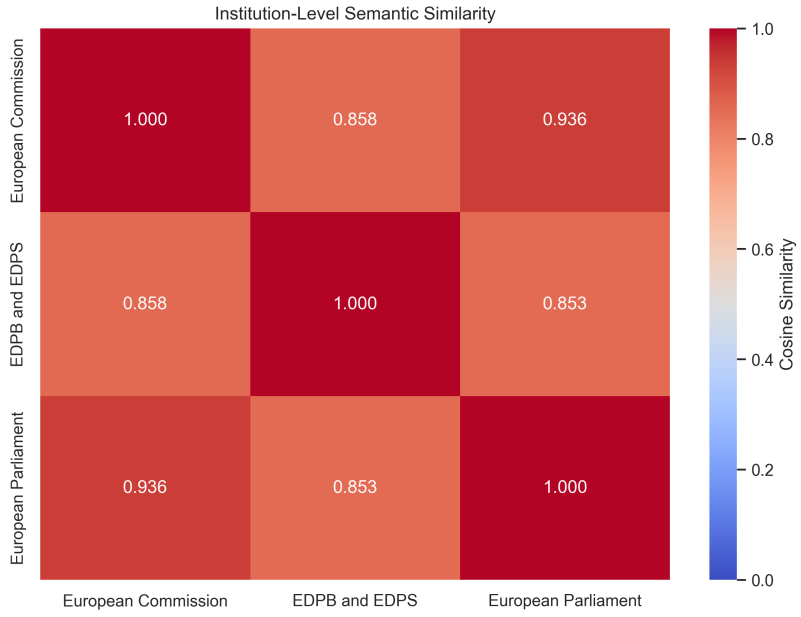

Figure 4: Institution-level semantic similarity based on cosine similarity of mean document embeddings. Values closer to 1 indicate stronger semantic alignment.

5.4 Amoore Axis Anchor Terms

Table 1 reports the exact anchor phrases used to construct the two poles of the Amoore axis in the embedding space. These terms define the semantic direction from legal norms/certainty to preemption/exception and were encoded exactly as written below.

Table 1: Anchor terms used for the Amoore axis construction

Preemption/Exception Pole	**Legal Norms/Certainty Pole**
detection, risk-based, automated suspicion, proactive monitoring, prevention, possible threat, indicators, algorithmic surveillance, preemptive intervention, identification	due process, proportionality, legal certainty, evidence-based, judicial oversight, fundamental rights, rule of law, probable cause, limitation, judicial warrant

Replication note. To reproduce axis construction exactly, use the same anchor terms, embedding model (`all-MiniLM-L6-v2`), and random seed (`numpy.random.seed(42)`).

5.5 Computational Reproducibility

Code repository: github.com/Joel-Arenz/governing_the_possible

Environment: Analyses were run on macOS with Python 3.14.2. Dependencies are listed in `requirements.txt`.

Randomization control: A fixed seed was used for downsampling (`numpy.random.seed(42)`), ensuring deterministic chunk selection under the same code/data setup.

Replication steps:

1. Clone repository and navigate to project root.
2. Create and activate virtual environment.
3. Install dependencies from `requirements.txt`.
4. Run `src/analysis.ipynb`